

ERA 5 calibration to Elexon power

S. Gomez¹

¹School of Mathematics, University of Edinburgh

2026-08-26

Table of contents I

- 1 Model validation framework
- 2 Out-of-sample analysis
- 3 Low wind speed events distribution
- 4 Next steps

Updates

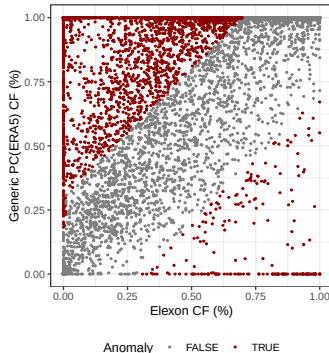
New

- Annual Review 2026
- Simulation for aggregation

Simulating aggregation with different anomaly thresholds

- WF level data contains anomalies consistent with curtailment or outages
- Previously only observations 50% away from Generic power curve estimate were excluded
- A simulation exercise now includes a 30% threshold

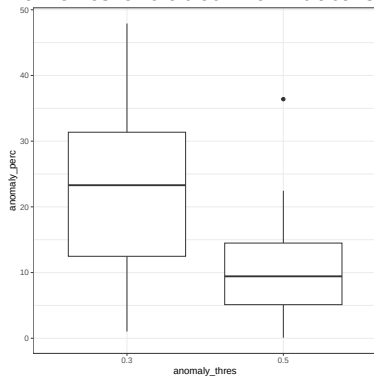
Anomalies excluded from data set



Anomaly exclusion

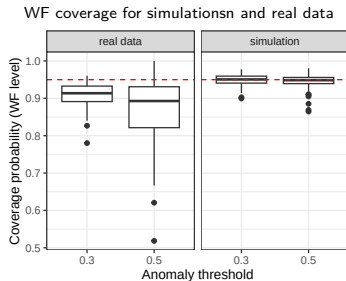
- WF level data contains anomalies consistent with curtailment or outages
- A 50% usually excluded less than 15% of the observations
- A 30% threshold is more aggressive and can exclude up to 30% of the observations

Anomalies excluded from data set

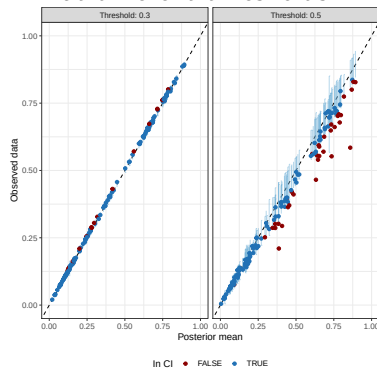


Effects for aggregated coverage

- Aggregation results in better coverage after excluding more anomalies
- Uncertainty bands seem quite narrow though



Posterior mean and 95% credible interval
at different thresholds



Annual Review

Project 2: Calibration of Reanalysis Using Wind Power Data

- Three approaches tried: aggregated GB model, power curve estimation (ERA5), deterministic power curve calibration
- Each excels at different aspect: country-level uncertainty, power curve shape, wf-level calibration
- Naive aggregation biases estimates, understates uncertainty (curtailment-like anomalies)
- **Next phase:**
 - Update uncertainty quantification for wf + aggregated calibration
 - Extend validation across wider range of periods/conditions
 - Model low-generation anomalies vs. exclude via threshold
 - Reassess low-generation events with updated estimates

Annual Review

Project 3: Extension of Statistical Calibration to Solar Generation

- Extend the wind calibration to solar generation
- Test wind benchmark models and validation framework
- Add solar-specific adjustments on the solar loss factor:
 - Account for diurnal cycles,
 - cloud coverage,
 - terrain effects
- **Goals:**
 - Obtain calibrated solar estimates matching the aggregated data
 - Characterise low-generation events for solar
 - Estimate the joint distribution of low-generation events
- Prioritised for months 6-10, pending Project 2 progress

Annual Review

Project 4: Impact of Low-Generation Events on Regional Demand and Transmission

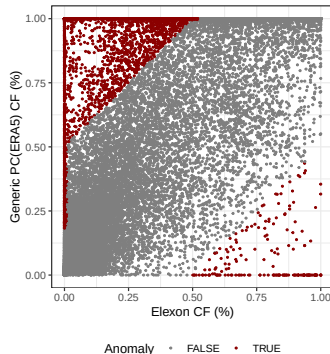
- Measure impact of low-generation events given energy demand requirements
- Build on the low-generation events distribution to assess demand shortfalls
- Construct a regional demand model using Elexon data, calibrated to observed national demand
- **Goals:**
 - Understand how low-generation events interact with demand dynamics and transmission constraints
 - Evaluate frequency and severity of worst-case scenarios, in terms of megawatt-hour gaps
- Tentative, months 10-12: full chapter, shorter chapter, or future work depending on progress

Model validation framework

Anomalies exclusion

- Data contains observations consistent with curtailment or outages
- These anomalies are excluded from the data set to avoid biasing the calibration
- Exclusions:
 - Elaxon close to 0% when ERA5 estimate is beyond mid regime
 - ERA5 estimate close to 100% when Elaxon is below mid regime
 - Difference greater than 50%

Anomalies excluded from data set



Updated models

- Wind farm level models
 - 1 Linear model (LM)
 - 2 Linear model with AR1 errors
 - 3 LM beta or t student errors
 - 4 Spatio-temporal model
 - 5 Quantile mapping (QM)
- Models on Great Britain aggregated power
 - 6 Linear model (GB LM)

Model equation

Linear model (LM)

$$\begin{aligned}\mathbb{E}(p_{it}^{\text{calib}}) &= \alpha_k + \beta_k \hat{p}_{it} + \gamma e_i + \delta d_i + s_{w,k}(w_{it}) + \epsilon_{it}, \\ \epsilon_{it} &\sim \mathcal{N}(0, \sigma_\epsilon)\end{aligned}\tag{1}$$

where the intercept and slope vary by type $k : \{\text{onshore, offshore}\}$, e_i and d_i represent the land elevation and the distance to the coast at location i , respectively, and $s_{w,k}(\cdot)$ is a smooth function that serves as a power regime correction.

Alternative models

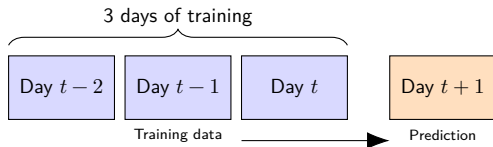
Spatiotemporal model (ST)

$$\begin{aligned}\mathbb{E}(p_{it}^{\text{calib}}) &= \alpha_k + \beta_k \hat{p}_{it} + \gamma e_i + \delta d_i + s_{w,k}(w_{it}) + u_t, \\ u_{it} &= \xi_i \cdot \tau_t.\end{aligned}\tag{4}$$

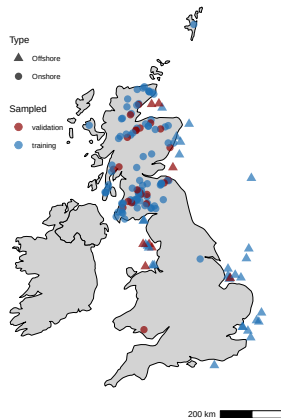
The additional term u_{it} is a separable spatiotemporal random effect. It consists of ξ_i , a zero-mean Gaussian spatial field over farm locations, and τ_t , a temporal component modelled as an AR(1) process.

Framework for OOS assessment

- Run models on a sample of days
 - 80/20 WF split for training and testing
 - Training with 7 days to predict the next 72 hours
 - Randomly chosen 15 days from 2025, 5 per regime



Sampled days for performance assessment

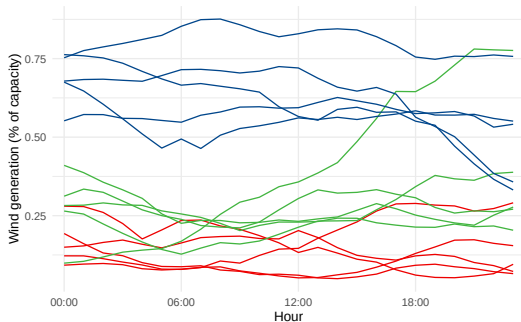


Sampled days in different regimes

- Objective is to assess performance by regime
- $t+1$ day are from specific regimes of wind speed
- Randomly chosen 5 days per regime
- Previous week, $t+2$, and $t+3$ can be anything

Sampled days for performance assessment

Daily generation time series for sampled days

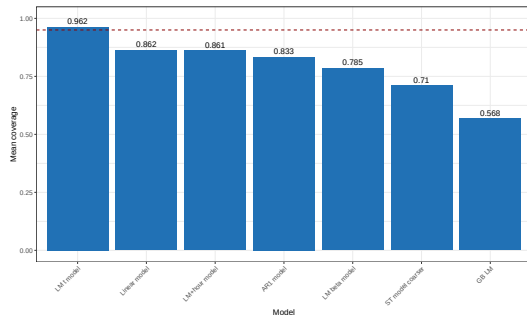


Key results

- Increasing model complexity improves mean power estimates
- However, uncertainty quantification for complex models misrepresents observed variability

Model	IS	OOS
ST model coarser	0.202	0.210
LM t model	0.212	0.208
QM	0.192	0.199
AR1 model	0.186	0.188
GB LM	0.180	0.185
Linear model	0.175	0.182
LM+hour model	0.176	0.181
LM beta model	0.175	0.181

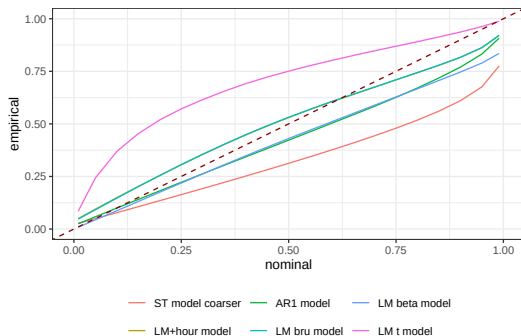
Coverage for day-ahead prediction bands



Reliability diagrams

- Linear model is the closest to true coverage, with some underdispersion in the upper tail
- LM Beta model and AR1 do better at lower tail but worse at upper tail

Coverage for day-ahead prediction bands



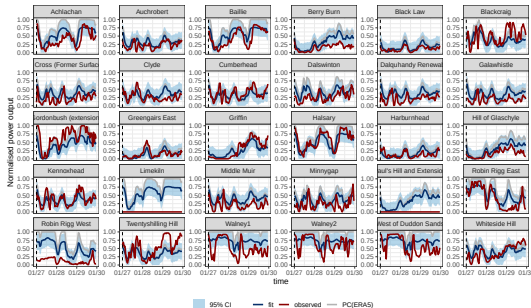
Probabilistic scores

Model	CRPS	Energy	Log	Brier score (CF threshold)		
				1%	5%	10%
LM beta model	0.086	11.426	-0.471	0.083	0.098	0.110
LM+hour model	0.094	11.665	0.558	0.097	0.115	0.124
LM bru model	0.094	11.685	0.564	0.098	0.115	0.124
AR1 model	0.102	12.295	0.681	0.121	0.151	0.161
ST model coarser	0.121	14.424	2.806	0.130	0.163	0.181
LM t model	0.169	36.843	0.153	0.131	0.143	0.148

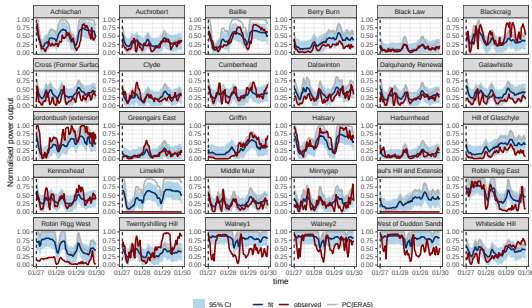
Out-of-sample analysis

Wind farm prediction bands (oos time)

Linear model (27-01-2025)

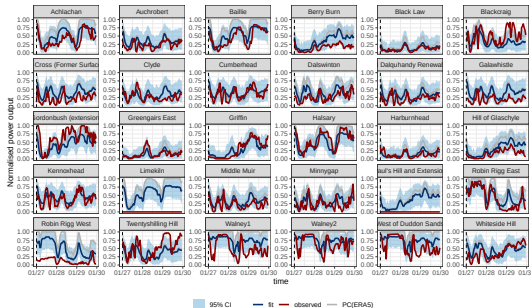


Spatiotemporal model coarse (27-01-2025)

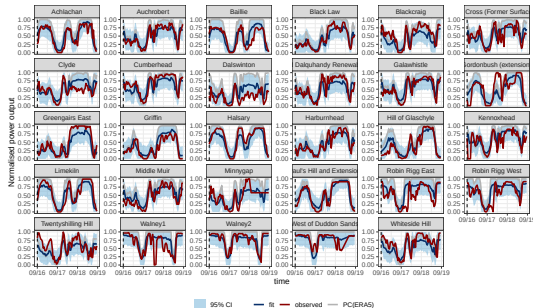


Wind farm prediction bands (oos time)

LM beta (27-01-2025)

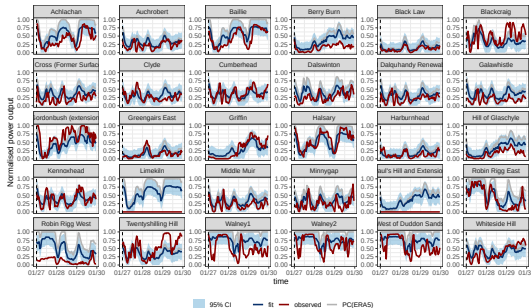


LM t-student (16-09-2025)

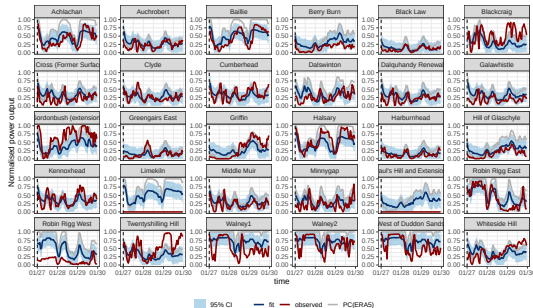


Wind farm prediction bands (oos time)

LM+hour model (27-01-2025)

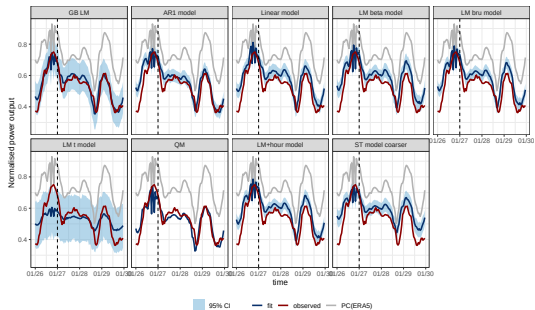


AR1 model (27-01-2025)

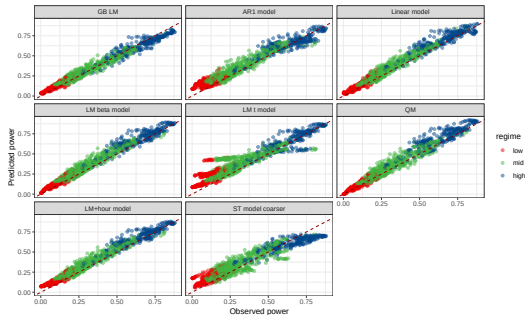


Aggregated prediction bands (oos time)

GB level



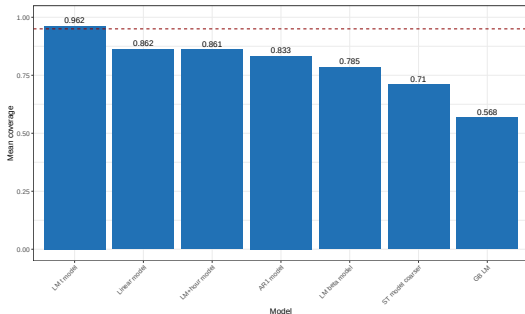
Scatter plot oos



Day-ahead forecast coverage (oos time)

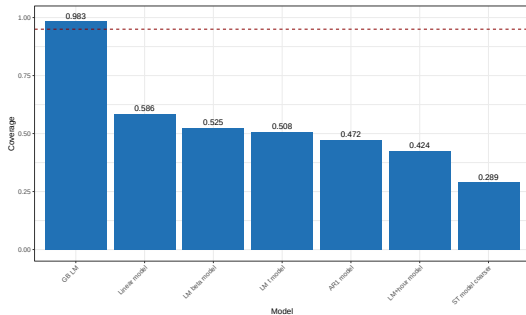
- Models at WF can achieve target coverage
- Aggregation reduces coverage
- High regime more susceptible to coverage loss
- Spatiotemporal model is not improving coverage

Coverage for day-ahead predictions

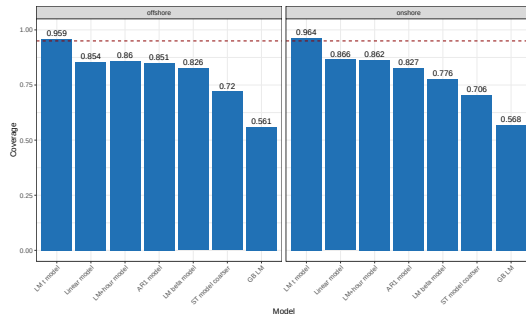


Coverage under aggregation (oos time)

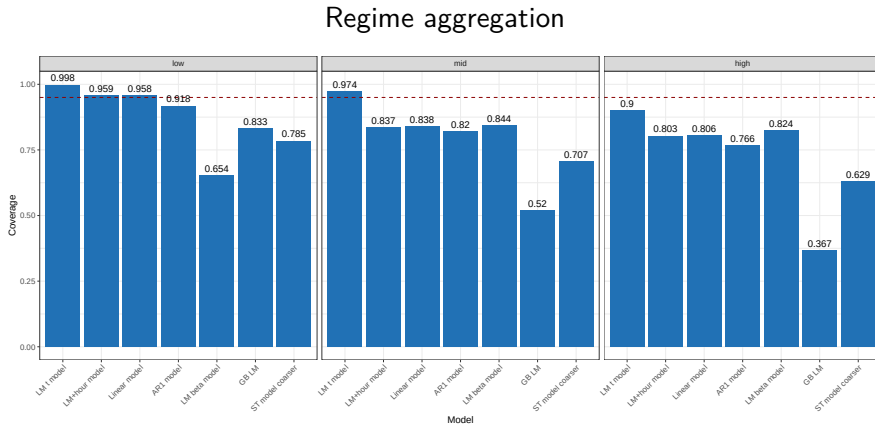
GB level



Onshore / offshore



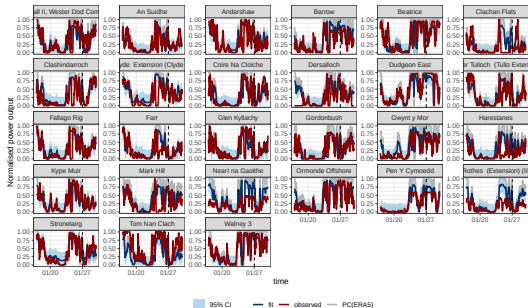
Coverage under aggregation by regime (oos time)



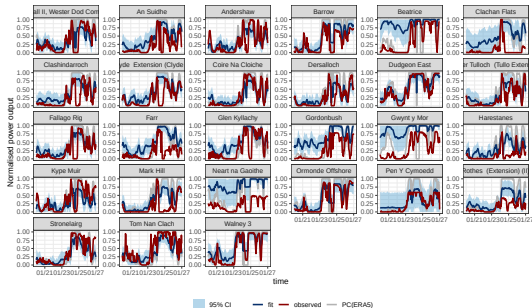
Out-of-sample locations

Wind farm prediction bands (oos space)

Linear model (27-01-2025)

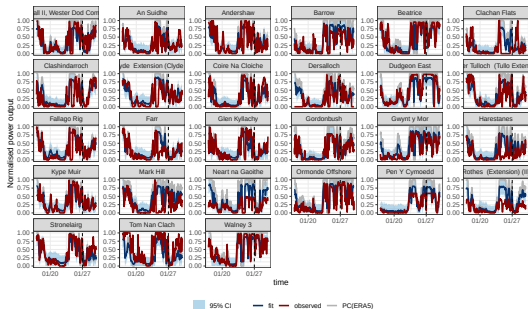


Spatiotemporal model coarse (27-01-2025)

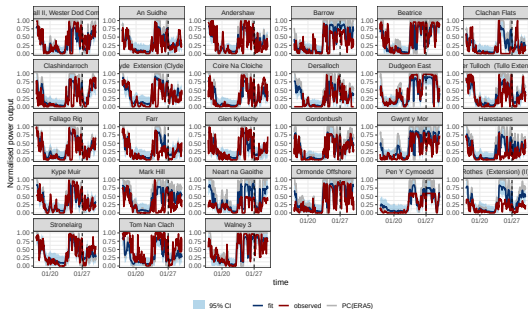


Wind farm prediction bands (oos space)

LM+hour model (27-01-2025)

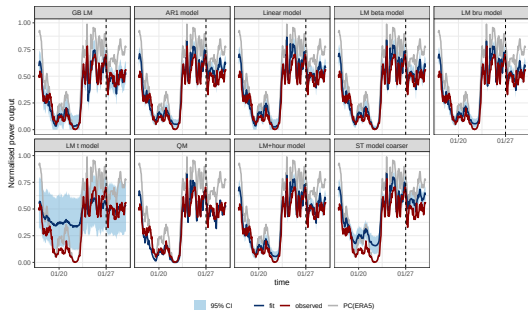


AR1 model (27-01-2025)

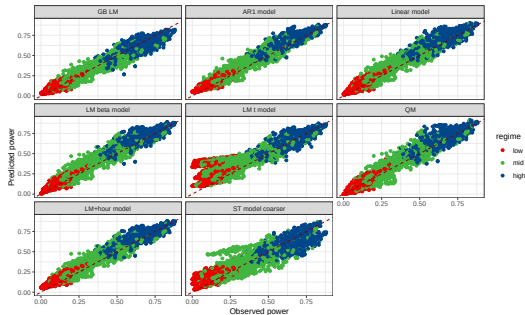


Aggregated prediction bands (oos space)

GB level (27-01-2025)

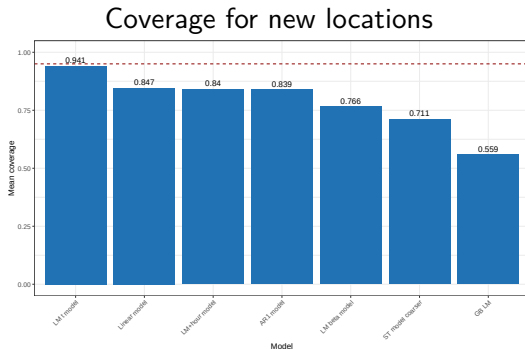


Scatter plot oos



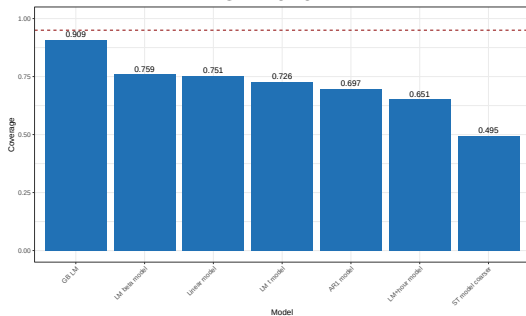
Unseen locations prediction intervals coverage (oos space)

- Attaining coverage in unseen locations is more challenging
- Isolated locations can lead to high uncertainty
- Location specific events may affect coverage
- Spatiotemporal model is not improving coverage

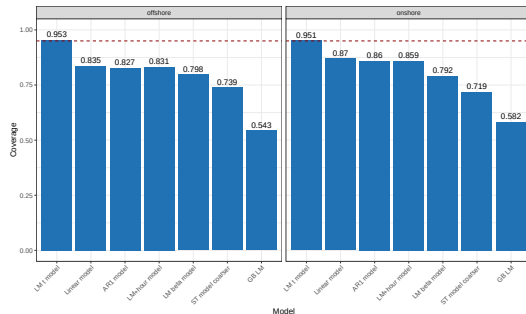


Coverage under aggregation (oos space)

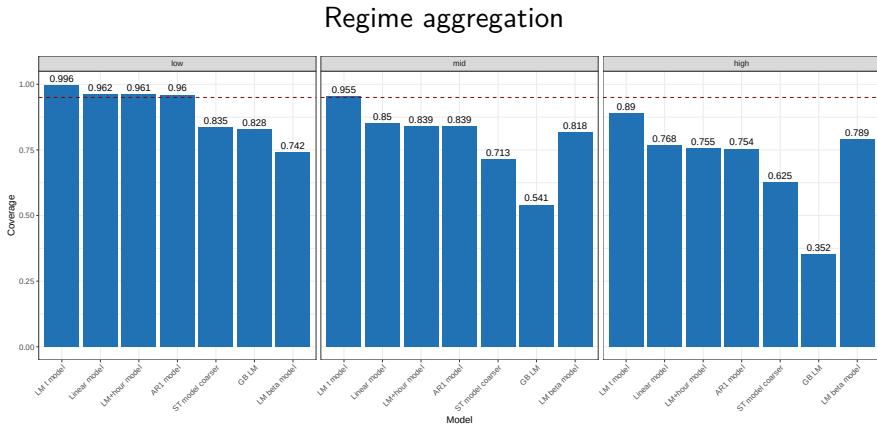
GB level



Onshore / offshore



Coverage under aggregation by regime (oos space)



Low wind speed events distribution

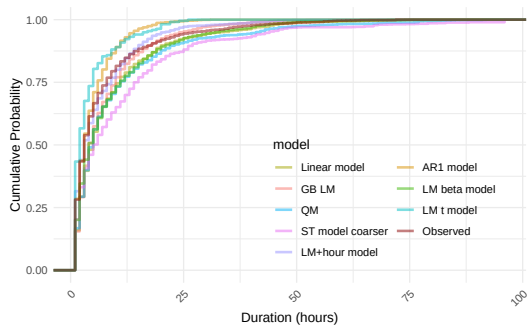
Low wind events (LWE) analysis

- Calibration should also consider LWE
- Using 0.05 as threshold for LWE
- Testing that model matches:
 - observed frequency of events
 - distribution of duration

Low wind speed events duration distribution

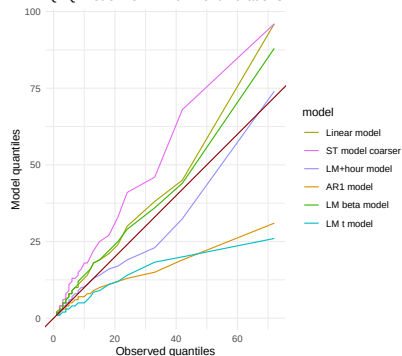
CDF

Cumulative Distribution of Low Wind Events Duration



QQ plot

Q-Q Plot of Low Wind Event Durations



Next steps

Addressing challenges

- Collapsing uncertainty under aggregation
- Spatiotemporal model is not improving calibration
- Low wind speed events are not well represented in the model samples

Appendix

Alternative models

Linear model with hour effect

This model results from adding a periodic hour effect h_t to control temporal effects

$$\mathbb{E}(p_{it}^{\text{calib}}) = \alpha_k + \beta_k \hat{p}_{it} + \gamma e_i + \delta d_i + s_{w,k}(w_{it}) + h_t + \epsilon_{it}, \quad (2)$$

Alternative models

Linear model with AR1 (or AR2) errors

A more robust temporal correlation term can be added to the model:

$$\begin{aligned}\mathbb{E}(p_{it}^{\text{calib}}) &= \alpha_k + \beta_k \hat{p}_{it} + \gamma e_i + \delta d_i + s_{w,k}(w_{it}) + u_t, \\ u_t &\sim \text{AR}(1),\end{aligned}\tag{3}$$

where u_t captures residual temporal autocorrelation, satisfying $u_t = \rho u_{t-1} + \varepsilon_{it}$ with $|\rho| < 1$

Alternative models

Quantile mapping (QM)

This approach doesn't include covariates but instead of correcting only the mean bias, makes the model match the distribution of observed power.

$$p_{t_k}^{\text{calib}} = F_{\text{obs}}\left(F_{\text{mod}}^{-1}(\hat{p}_{t_k})\right), \quad (5)$$

where F_{mod} is the empirical CDF of the ERA5-derived power estimate and F_{obs} is the empirical CDF of the observed generation. The transfer function $F_{\text{obs}} \circ F_{\text{mod}}^{-1}$ is a non-decreasing smooth function that corrects distributional bias by power regime.

Alternative models

Linear model (GB LM)

If the goal is only to produce estimates at the aggregate level, the optimal correction is achieved by modelling aggregate time series instead of aggregating estimates.

$$p_t = \sum_i p_{it}$$

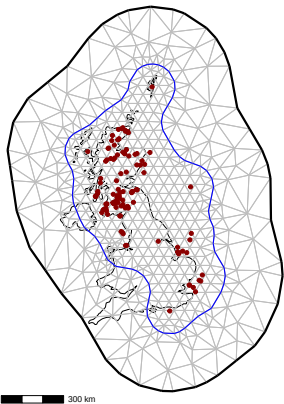
$$\mathbb{E}(p_t^{\text{calib}}) = \alpha_k + \beta_k \hat{p}_t + s_{w,k}(\tilde{w}_t) + \epsilon_t, \quad (4)$$

$$\epsilon_t \sim \mathcal{N}(0, \tilde{\sigma}_\epsilon), \quad (6)$$

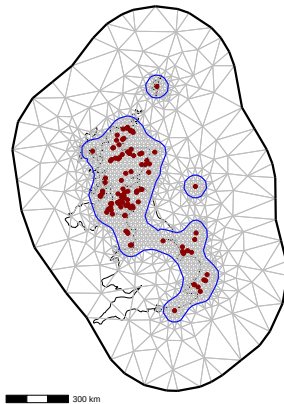
where now wind power p_t is the average capacity factor of all the wind farms, $\tilde{w}_t$ is the average wind speed of the observed locations, and ϵ now represents the error in aggregated terms.

Meshes for spatiotemporal models

Coarse mesh



Fine mesh



Error metrics and performance assessment

Error metrics

Model	RMSE		MAE		MDAPE (%)		Bias	
	IS	OOS	IS	OOS	IS	OOS	IS	OOS
GB LM	0.212	0.203	0.147	0.138	31.8	36.7	0.024	0.047
AR1 model	0.184	0.193	0.135	0.137	30.4	37.2	0.010	0.055
AR2 model	0.184	0.194	0.135	0.137	30.4	37.3	0.010	0.056
Linear model	0.189	0.188	0.132	0.130	31.9	36.5	0.001	0.033
LM+hour model	0.061	0.193	0.041	0.134	9.8	41.7	0.004	0.041
Spatio-temporal coarse	0.104	0.182	0.070	0.124	16.5	40.2	0.005	0.032
Spatio-temporal fine	0.071	0.171	0.046	0.113	10.0	37.2	0.003	0.042

Error metrics

model	RMSE	MAE	Bias
QM	0.210	0.135	1.58e-04
GB LM	0.191	0.130	2.12e-02
Linear model	0.168	0.117	1.76e-03
1D SPDE model	0.167	0.116	1.46e-03
AR2 model	0.164	0.113	1.45e-03
AR1 model	0.164	0.113	1.44e-03
Spatio-temporal coarse	0.124	0.082	1.26e-03
Spatio-temporal fine	0.097	0.061	1.16e-03

Error metrics by regime

Model	Low			Mid			High		
	RMSE	Bias	MAE	RMSE	Bias	MAE	RMSE	Bias	MAE
QM	0.110	0.062	-1.59e-03	0.196	0.133	1.08e-04	0.305	0.219	2.21e-03
GB LM	0.108	0.069	2.05e-02	0.186	0.132	2.44e-02	0.262	0.193	1.44e-02
1D SPDE model	0.093	0.059	7.34e-03	0.160	0.115	2.50e-03	0.234	0.179	-7.56e-03
Linear model	0.092	0.059	6.57e-03	0.162	0.117	2.26e-03	0.235	0.183	-4.75e-03
AR2 model	0.089	0.056	3.07e-03	0.156	0.112	1.24e-03	0.230	0.176	1.49e-04
AR1 model	0.089	0.056	3.05e-03	0.156	0.112	1.22e-03	0.230	0.176	1.85e-04
Spatio-temporal coarse	0.061	0.037	2.22e-03	0.117	0.081	1.13e-03	0.180	0.132	5.07e-04
Spatio-temporal fine	0.048	0.028	1.75e-03	0.088	0.059	1.05e-03	0.145	0.101	7.84e-04

Error metrics by technology

Model	Offshore			Onshore		
	RMSE	Bias	MAE	RMSE	Bias	MAE
QM	0.213	0.139	5.85e-02	0.209	0.133	-1.85e-02
GB LM	0.184	0.127	8.26e-03	0.193	0.131	2.54e-02
1D SPDE model	0.172	0.119	6.88e-04	0.166	0.114	1.70e-03
AR1 model	0.169	0.117	6.72e-04	0.162	0.111	1.68e-03
AR2 model	0.169	0.117	7.36e-04	0.162	0.111	1.67e-03
Linear model	0.157	0.113	2.84e-03	0.172	0.118	1.42e-03
Spatio-temporal coarse	0.098	0.066	1.26e-03	0.131	0.087	1.26e-03
Spatio-temporal fine	0.073	0.048	5.82e-04	0.103	0.065	1.34e-03

References

- Cannon, D. J., D. J. Brayshaw, J. Methven, P. J. Coker, and D. Lenaghan. 2015. "Using Reanalysis Data to Quantify Extreme Wind Power Generation Statistics: A 33 Year Case Study in Great Britain." *Renewable Energy* 75 (March): 767–78. <https://centaur.reading.ac.uk/38448/>.
- Potisomporn, P. 2024. "Spatiotemporal Variability of Extreme Low Wind Power Events in Great Britain - ORA - Oxford University Research Archive." [Http://purl.org/dc/dcmitype/Text](http://purl.org/dc/dcmitype/Text).
<https://ora.ox.ac.uk/objects/uuid:de37bee2-1813-42e5-a69c-9b15f038d096>.
- Thornton, Hazel E. 2020. "Atmospheric Circulation, Seasonal Predictability and Britain's Energy System." PhD thesis, University of Reading.
<https://doi.org/10.48683/1926.00095832>.